Biology-informed reinterpretation of lattice radiotherapy using non-local bystander signalling and anatomy-aware vascular sink modelling

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Abstract

Objective. Lattice radiotherapy is usually evaluated using physical dose metrics such as target coverage, peak-to-valley dose ratio (PVDR) and organ-at-risk dose. These metrics do not explicitly represent non-local biological signalling between high-dose vertices and low-dose valleys. This study developed a hypothesis-generating computational framework combining a transferable spatiotemporal bystander model with a synthetic head-and-neck lattice cohort to test whether physically plausible plans retain the same risk interpretation after non-local signalling and anatomy-aware vascular uptake are considered.

Approach. A reaction–diffusion framework propagated short-range reactive oxygen species-like and longer-range cytokine-like signals from Monte Carlo-derived dose distributions. The transport model was calibrated once in a one-dimensional half-field assay and transferred without parameter retuning into two-dimensional stripe, three-dimensional lattice and irregular surrogate geometries. A fully synthetic 10-case bulky head-and-neck cohort was then generated with tumour, organ-at-risk and vessel masks. TOPAS was used to generate two-component photon lattice dose distributions comprising broad peripheral coverage and focal vertex boosts. Each case was rescored as physical-only, biology-informed without vascular uptake and biology-informed with anatomical vascular uptake. Repeat-run uncertainty and vascular-sink falsification tests were performed.

Main results. The locked model predicted persistent biological valley burden despite physical valley sparing. In the irregular surrogate plan, valley equivalent uniform dose increased from 0.218 Gy under linear-quadratic interpretation to 2.831 Gy under the full model. In the 10-case cohort, biological reinterpretation changed case ranking in 8/10 cases and produced model-derived brainstem-adjacent burden flags in 5/10 cases. Mean PVDR decreased from 1.817 physically to 1.478 without vascular uptake and 1.436 with anatomical uptake. Sink-related endpoint shifts exceeded the combined uncertainty band in 40/42 comparisons, while robust sink-driven rank reversals remained uncommon.

Significance. This work is a hypothesis-generating computational risk-analysis framework, not a validated clinical toxicity predictor or treatment-planning optimizer. It provides a reproducible method for testing whether physically acceptable lattice dose patterns may acquire different biological interpretations when non-local signalling and anatomy-aware vascular uptake are included. Translation requires higher-statistics transport, patient-derived RTDOSE/RTSTRUCT data and experimental SFRT assay validation before any clinical decision-support use.

Keywords: spatially fractionated radiotherapy; lattice radiotherapy; TOPAS; bystander effect; vascular sink; biological modelling; head and neck radiotherapy

1 Introduction

Spatially fractionated radiation therapy (SFRT) challenges the conventional assumption that therapeutic radiation should be delivered as a spatially uniform dose across the target. Instead, SFRT deliberately creates heterogeneous dose distributions containing high-dose peak regions and lower-dose valley regions. This family includes GRID therapy, microbeam radiotherapy, minibeam radiotherapy and three-dimensional lattice radiotherapy. Interest in SFRT has increased because spatial modulation may permit dose escalation in bulky, radioresistant, recurrent or anatomically constrained tumours while reducing normal-tissue exposure Prezado et al. (2024); Prezado (2022); Fernandez-Palomo et al. (2022).

Lattice radiotherapy is a modern three-dimensional SFRT implementation in which discrete high-dose vertices are placed inside the tumour while a lower-dose background or peripheral component maintains broader target coverage. Contemporary lattice stereotactic body radiotherapy workflows can be delivered using image-guided photon planning systems, and the literature has consequently focused on vertex diameter, spacing, placement, prescription strategy, PVDR, target coverage, dose fall-off and organ-at-risk sparing Duriseti et al. (2021); Kavanaugh et al. (2022); Duriseti et al. (2022); Grams et al. (2023); Iori et al. (2023); Zhang et al. (2023); Botti et al. (2024); Yang et al. (2022).

Although these physical metrics are essential, they provide an incomplete description of highly heterogeneous irradiation. A lattice plan creates steep gradients in which heavily irradiated subvolumes sit adjacent to lower-dose valleys. These valleys are often interpreted dosimetrically, but they should not necessarily be regarded as biologically passive. Preclinical SFRT reviews have questioned whether peak dose alone is an adequate response descriptor, emphasizing that valley dose, spatial scale and tissue-level response can be critical determinants of outcome Fernandez-Palomo et al. (2022); Prezado et al. (2024).

A central reason physical dose alone may be insufficient is that radiation response is not purely local. Radiation-induced bystander effects describe responses in cells receiving little or no direct dose but affected by signals from irradiated cells. These signals may involve reactive oxygen species, inflammatory cytokines, DNA-damage mediators, oxidative metabolism, gap junction communication and soluble stress-response factors Prise and O'Sullivan (2009); Seymour and Mothersill (2004); Azzam et al. (2003); Mothersill and Seymour (2003); Blyth and Sykes (2011). In SFRT, where peaks and valleys coexist over short distances, such non-local signalling may alter survival, inflammatory burden and normal-tissue response in ways not captured by conventional dose-volume histograms.

Mathematical and computational models provide a mechanism for testing this hypothesis. Prior work has separated direct radiation injury from bystander-mediated response and shown that heterogeneous or gradient irradiation can produce outcomes that differ from local dose-response models alone Ebert et al. (2010); McMahon et al. (2013); Peng et al. (2018); Powathil et al. (2016). SFRT-specific modelling has further shown that bystander effects can alter equivalent uniform dose, mean biological dose and peak-valley interpretation Cahoon et al. (2021); Mahmoudi et al. (2022). These studies support treating SFRT plan assessment as more than a physical-dose problem.

Most signalling-based SFRT interpretations, however, treat signal decay or removal as generic rather than anatomy-specific. This is limiting because the tumour microenvironment is spatially structured. Blood vessels influence oxygenation, perfusion, endothelial response, immune-cell trafficking and clearance of soluble mediators. In microbeam and related spatially fractionated modalities, vascular response and redistribution have been proposed as contributors to differential tumour and normal-tissue effects Bouchet et al. (2015); Ortiz and Ramos-Méndez (2024). SFRT has also been linked to immune and microenvironmental mechanisms beyond direct DNA damage Lu et al. (2024). Vasculature may therefore act not only as avoidance anatomy, but also as a spatial modifier of propagated biological stress.

This raises an unresolved question for lattice radiotherapy: if non-local bystander signalling is incorporated into plan evaluation, does anatomical vasculature measurably modify the resulting biological burden? A vessel-associated uptake term could represent local clearance, washout, binding or removal of diffusible stress mediators. Such a term is meaningful only if its effect depends on vessel location relative to tumour, high-dose vertices, valley regions and nearby organs at risk. If the same effect can be reproduced by uniform washout or arbitrary smoothing, anatomical vascular uptake adds little mechanistic information.

The present study addresses this gap by developing a biological risk-analysis framework combining Monte Carlo-derived physical dose, non-local bystander signalling and anatomy-aware vascular uptake. TOPAS and Geant4 provide the transport-grounded physical dose basis Agostinelli et al. (2003); Perl et al. (2012), while the biological layer converts dose into survival and effective-dose-like maps using linear-quadratic and equivalent-uniform-dose concepts Fowler (1989); Niemierko (1997). The central novelty is not simply adding a vascular sink, but falsification-testing it against no-sink, uniform-washout, spatially perturbed and vessel-dropout controls. We hypothesized that physically plausible lattice plans would not remain biologically equivalent after non-local signalling, and that anatomy-aware vascular uptake would behave as a measurable secondary modifier rather than a generic washout correction.

2 Methods

2.1 Study design and workflow

The workflow is summarized in figure 1. The study had four linked stages: (i) calibration and transfer testing of a non-local biological model, (ii) construction of synthetic head-and-neck lattice cases, (iii) TOPAS-based physical dose generation and (iv) biological reinterpretation with uncertainty and vascular-sink falsification. The model was used as a post-processing risk-analysis layer applied to fixed physical dose distributions. It was not used as a clinical optimizer and did not generate patient-specific treatment recommendations.

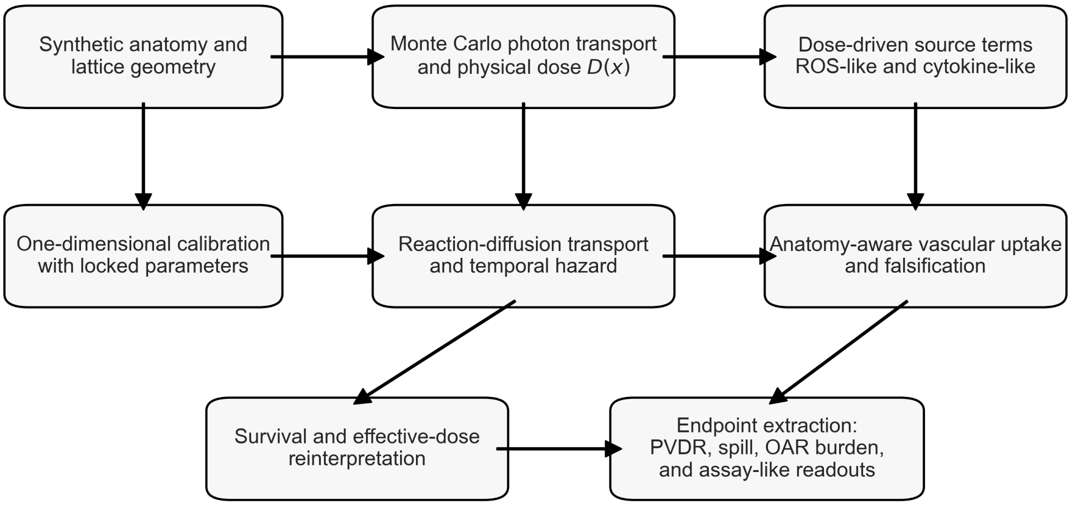

Figure 1: Integrated workflow. Synthetic lattice geometry was combined with Monte Carlo photon transport and biology-informed post-processing to derive survival-based reinterpretation, anatomy-aware vascular uptake effects, and endpoint-level risk metrics.

2.2 Transferable non-local biological model

Two signaling species were modeled: a short-range reactive oxygen species-like channel and a longer-range cytokines-like channel. For species k, concentration $C_k(\mathbf{x}, t)$ evolved as

$$\frac{\partial C_k}{\partial t} = D_k \nabla^2 C_k - \lambda_k C_k - u_k(\mathbf{x}) C_k + E_k(\mathbf{x}), \tag{1}$$

where D_k is the diffusion coefficient, λ_k is the decay term, $u_k(\mathbf{x})$ is spatial uptake and $E_k(\mathbf{x})$ is the dose-driven emission source. Emission was represented as

$$E_k(\mathbf{x}) = E_{\max,k} \left[1 - \exp\{-\gamma D_{\text{phys}}(\mathbf{x})\}\right] M_{\text{type}}(\mathbf{x}, k) M_{\text{oxygen}}(\mathbf{x}, k), \tag{2}$$

where γ controls transition into the signalling regime. In no-sink simulations, $u_k(\mathbf{x}) = 0$. In anatomical-sink simulations, $u_k(\mathbf{x})$ was non-zero inside vessel masks, with stronger uptake assigned to the cytokine-like channel than to the short-lived ROS-like channel.

Instantaneous stress was integrated over time,

$$H(\mathbf{x}) = \int_0^{t_{\text{final}}} \left[w_{\text{ROS}} C_{\text{ROS}}(\mathbf{x}, t) + w_{\text{cyto}} C_{\text{cyto}}(\mathbf{x}, t)\right] dt, \tag{3}$$

with $w_{\text{ROS}} = w_{\text{cyto}} = 0.4$. Direct radiation response used

$$S_{\text{LQ}}(\mathbf{x}) = \exp[-\alpha D(\mathbf{x}) - \beta D(\mathbf{x})^2], \tag{4}$$

and final survival was

$$S_{\text{final}}(\mathbf{x}) = S_{\text{LQ}}(\mathbf{x}) \exp[-s(H(\mathbf{x}) + w_{\text{immune}} P_{\text{immune}})], \tag{5}$$

where s is the non-local coupling coefficient and $w_{\text{immune}} = 0.2$. Survival was inverted to an effective-dose-like quantity,

$$D_{\text{eff}}(\mathbf{x}) = \frac{-\alpha + \sqrt{\alpha^2 - 4\beta \ln S_{\text{final}}(\mathbf{x})}}{2\beta}. \tag{6}$$

Calibration was performed once in a one-dimensional half-field assay geometry, targeting survival of 0.70 at +2 mm outside the field, below 0.01 at -2 mm inside the field and approximately 0.95 at +10 mm in the shielded region. After calibration, $D_{\text{cyto}} = 1.2$, $\lambda_{\text{cyto}} = 0.001$, $\gamma = 0.35$ and $s = 0.0029365$ were locked and transferred without retuning to two-dimensional stripe, three-dimensional lattice and irregular surrogate-plan geometries (figure 2).

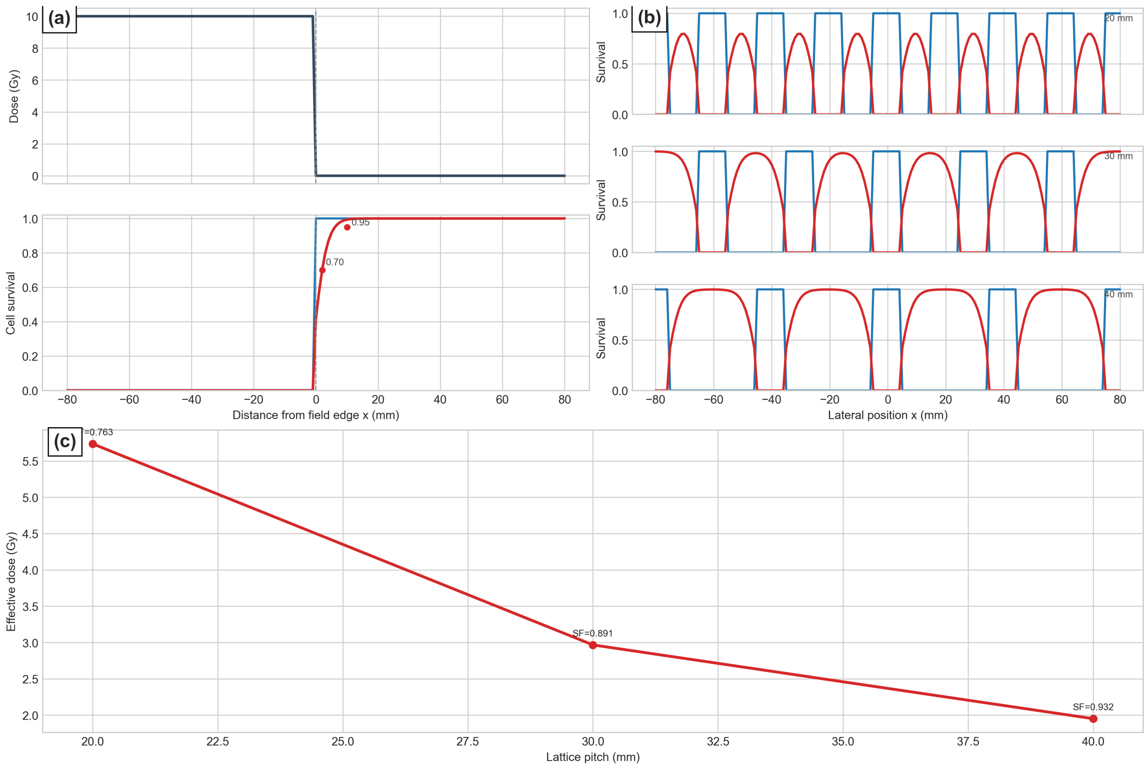

Figure 2: Calibration and transferability of the locked biological model. (a) One-dimensional half-field calibration used to lock the biological parameter set. (b) Stripe-pattern transfer tests across representative lattice spacings. (c) Holdout three-dimensional validation showing the effective-dose response as a function of lattice pitch.

2.3 Synthetic head-and-neck cohort and physical dose generation

A fully synthetic 10-case head-and-neck cohort was generated using algorithmic tumour, organ-at-risk and vessel masks (figure 3). Sites included sinonasal/maxillary, oropharyngeal, laryngo-hypopharyngeal, parapharyngeal, buccal, oral tongue/floor-of-mouth, nodal, deep parotid/infratemporal and composite upper-neck presentations. Estimated gross tumour volumes ranged from approximately 66 to 326 cm^3. The cohort contained no patient-identifiable data.

All cases followed a common lattice boost concept. The peripheral lattice target was defined as $\mathrm{CTV_{boost}} = \mathrm{GTV} + 5$ mm. Vertices were assigned a nominal 15 Gy peak concept and the peripheral target received 3.0–3.5 Gy. Candidate vertices were accepted only if they remained intratumoural, preserved spacing and avoided high-priority anatomy. The fixed safe-core branch used a 5 mm contracted GTV, 15 mm OAR clearance, 6 mm candidate sampling step, 6 mm candidate support radius and 18 mm minimum inter-vertex spacing.

TOPAS was used as the physical dose engine. Each case was represented as a transport-ready `TsImageCube` phantom. Rather than reproducing a full clinical VMAT sequence, dose was generated with a two-component photon lattice surrogate comprising a broad base component and a focused vertex component. Because absorbed dose is linear in fluence, the final physical dose was formed as

$$D_{\mathrm{phys}}(\mathbf{x}) = w_{\mathrm{base}} D_{\mathrm{base}}(\mathbf{x}) + w_{\mathrm{vert}} D_{\mathrm{vert}}(\mathbf{x}), \tag{7}$$

with weights chosen to match PTV $D_{95} \approx 3.5$ Gy and peak mean dose near 15 Gy. Default history counts were 120,000 for the base component and 240,000 for the vertex component.

Monte Carlo uncertainty was treated as a limitation and propagated into endpoint interpretation. The modest history counts enabled repeated synthetic-case analysis but do not constitute final clinical-quality transport. To reduce the risk that biological effects reflected stochastic noise alone, repeated seed/history-scale runs were performed in a six-case subset, TOPAS standard-deviation outputs were retained where available, and sink-driven endpoint shifts were compared with combined 95% Monte Carlo plus uptake-sensitivity bands. Body-aware Gaussian denoising was applied only after raw transport inspection and was used to stabilize macroscopic lattice metrics while preserving peak-valley structure. Higher-statistics transport, pre/post-denoising uncertainty maps and dose-grid convergence testing are required before clinical translation.

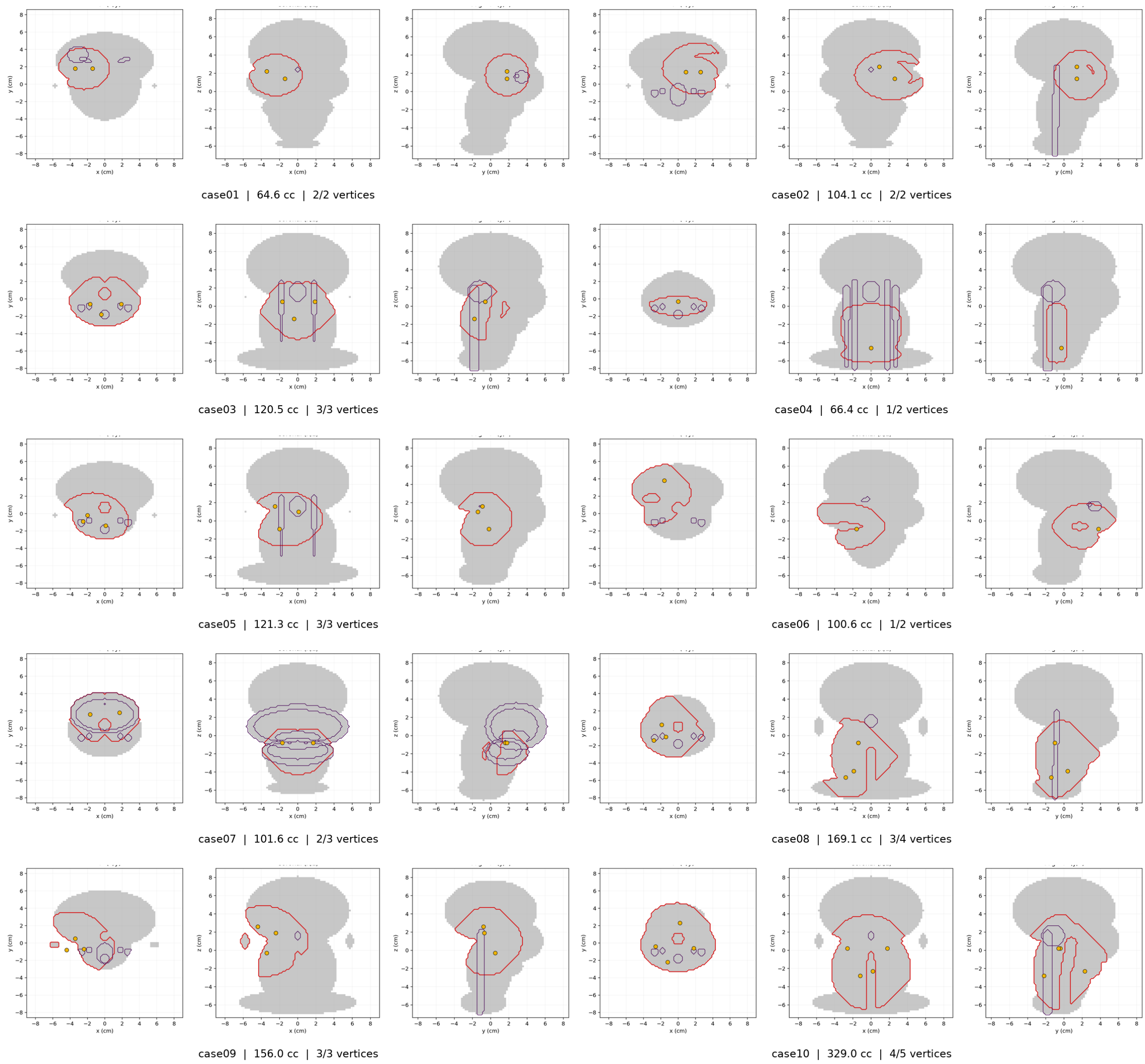

Figure 3: Synthetic head-and-neck cohort used for the site-specific risk-analysis branch. Ten benchmark lattice cases spanning distinct tumour sites, volumes, and final accepted vertex layouts after application of the geometric and organ-at-risk exclusion rules.

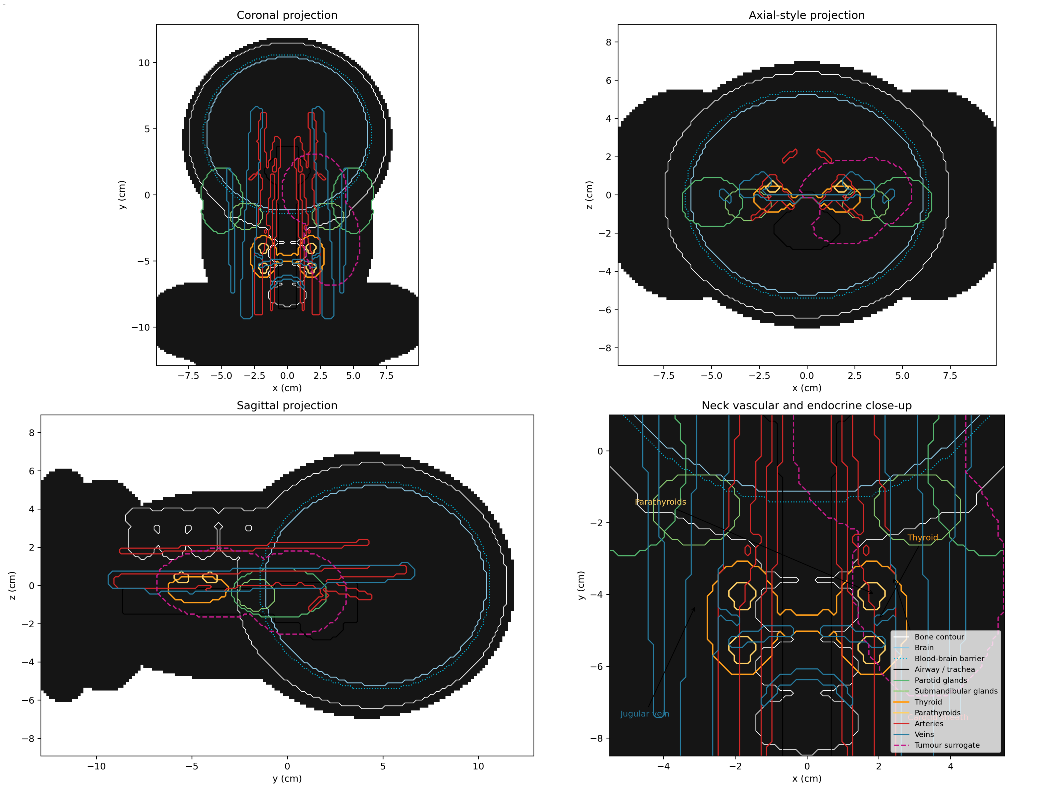

Figure 4: Synthetic head-and-neck reference anatomy. Coronal, axial, and sagittal views of the voxelized phantom showing the tumour surrogate, major organs at risk, and vascular-endocrine structures used for geometry definition and biology-aware post-processing

2.4 Endpoint extraction, terminology and falsification

Each case was processed in three modes: physical-only, biology-informed without vascular uptake and biology-informed with anatomical vascular uptake. Primary endpoints were PTV D_{95}, PVDR, peri-GTV 0–5 mm mean, peri-GTV 5–15 mm mean, spinal cord D_2, brainstem D_2 and right parotid mean. Outputs were separated into absorbed dose, effective-dose-like biological scores and relative assay-like proxies (table 1) to avoid implying that model-derived quantities are delivered dose or validated toxicity probability.

Table 1: Interpretation of physical, biological and assay-like outputs.

Output class	Examples	Interpretation
Absorbed physical dose	D_{phys}, PTV D_{95}, peak dose, valley dose, OAR D_2, PVDR	Monte Carlo-derived absorbed dose in Gy; describes the physical lattice distribution only.
Effective-dose-like biological score	D_{eff}, biological valley EUD, model-derived OAR burden score	Gy-like score obtained by LQ inversion of modelled survival; not delivered dose and not a validated toxicity predictor.
Relative assay-like proxy	γH2AX-like signal, TUNEL-like signal, cytokine AUC	Dimensionless or arbitrary-unit outputs used to express experimentally recognizable patterns; not calibrated assay measurements.

Repeat-run uncertainty used three seeds and three history scales in a representative six-case subset. A sink-driven endpoint shift was considered noise-qualified only when it exceeded the combined 95% Monte Carlo plus uptake-sensitivity band. The anatomical sink was challenged against no sink, mass-matched uniform washout, local vessel dropout within 20 mm of tumour, randomized anatomy-constrained displacement, superior-inferior flip and anterior-posterior flip. These controls tested whether the vascular-sink signal depended on anatomical placement rather than generic smoothing or global washout.

3 Results

3.1 Physical lattice-dose benchmark and plausibility check

Before biological reinterpretation, the physical lattice dose distributions were evaluated for consistency with published lattice and SFRT planning principles. This was a benchmark and plausibility check, not a claim of clinical plan equivalence. Published workflows commonly use intratumoural high-dose vertices within a lower-dose background, evaluated by target coverage, vertex dose, peak-valley contrast, dose fall-off and OAR avoidance Wu et al. (2020); Duriseti et al. (2021); Kavanaugh et al. (2022); Duriseti et al. (2022); Grams et al. (2023); Iori et al. (2023). The synthetic plans reproduced these broad design features: PTV $D_{95} \approx 3.5$ Gy by construction, intratumoural boosts near a nominal 15 Gy peak concept, 2–5 accepted vertices depending on tumour size and exclusions, and a cohort-average physical PVDR of 1.817. This modulation is more conservative than microbeam/minibeam peak-valley separation, but plausible for macroscopic photon lattice surrogates. The plans should therefore be interpreted as controlled, transport-grounded substrates for hypothesis generation, not clinically optimized VMAT, SBRT, proton or carbon-ion plans.

3.2 Locked non-local biology amplified low-dose valleys across geometries

The half-field calibration reproduced the intended survival pattern: 30% out-of-field bystander reduction at +2 mm, strong in-field killing and recovery toward unity by +10 mm. When transferred into stripe holdouts, centre-valley survival increased monotonically with pitch: 0.699 at 20 mm, 0.900 at 30 mm and 0.972 at 40 mm. In three-dimensional lattice holdouts sampled at 5.03 cm depth, the 20 mm pitch case had a physical valley dose of 0.668 Gy but an effective dose of 5.74 Gy; the 30 mm case had 0.306 Gy physically and 2.97 Gy effectively; and the 40 mm case had 0.184 Gy physically and 1.95 Gy effectively. Thus, increasing pitch improved sparing, but physical valley dose did not equal biological valley burden.

In the irregular surrogate plan, non-local biology broadened the response beyond the physical peak layout. At the sampled centre voxel, physical dose was 2.90 Gy and LQ-only survival was 0.894. After spatial transport, temporal hazard integration and the systemic term, survival fell to 0.595, corresponding to an effective dose of 9.07 Gy. Valley equivalent uniform dose increased from 0.218 Gy under LQ-only interpretation to 2.831 Gy under the full model, a +2.613 Gy non-local shift.

3.3 Biological reinterpretation changed cohort-level plan interpretation

All 10 site-specific synthetic cases completed the TOPAS workflow. Physical target coverage was stable by construction, with mean PTV $D_{95} = 3.50$ Gy, while physical PVDR and OAR-adjacent dose varied between cases. Biological rescoring substantially changed the endpoint profile (figure 5, table 2). Mean PVDR decreased from 1.817 physically to 1.478 without vascular uptake and 1.436 with anatomical uptake, indicating biological filling-in of the physical peak-valley pattern. Peri-GTV 0–5 mm mean increased from 7.219 Gy physically to 20.616 Gy without sink and 19.929 Gy with anatomical sink. The brainstem-adjacent D_2 score increased from 11.454 Gy physically to 28.399 Gy without sink and 27.951 Gy with sink; these values are model-derived burden scores after biological reinterpretation, not toxicity predictions or delivered absorbed dose.

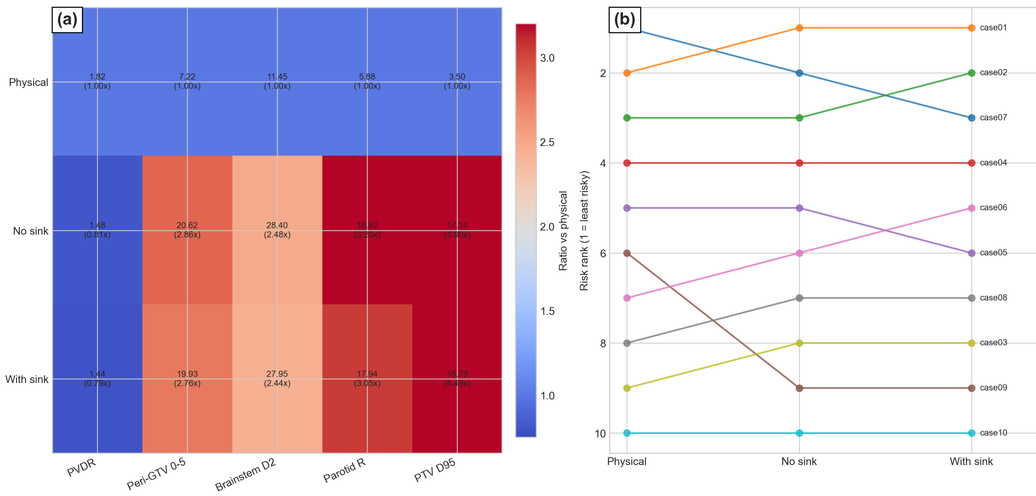

Figure 5: Biological reinterpretation of the 10-case cohort.(a) Cohort-mean endpoint ratios relative to the physical-only analysis, showing the change in PVDR, peri-tumour spill, brainstem burden, parotid burden, and target coverage after biology-informed post-processing. (b) Case-wise rank shifts across physical-only, no-sink, and anatomical-sink conditions, illustrating how biological modelling altered comparative plan interpretation.

Table 2: Cohort-mean endpoints across physical-only, no-sink and anatomical-sink modes. Values are absorbed dose in the physical-only column and effective-dose-like model scores in the biological columns, except for PVDR.

Endpoint	Physical-only	No sink	Anatomical sink
PTV D_{95} (Gy)	3.500	17.144	15.731
PVDR	1.817	1.478	1.436
Peri-GTV 0–5 mm mean (Gy)	7.219	20.616	19.929
Peri-GTV 5–15 mm mean (Gy)	5.923	16.900	16.492
Brainstem-adjacent D_2 score (Gy-like)	11.454	28.399	27.951
Right parotid mean (Gy-like)	5.875	18.821	17.939

Biology-informed analysis changed case ranking in 8/10 cases relative to physical-only ordering. The best physical case, an oral tongue/floor-of-mouth horseshoe geometry, fell to third when anatomical uptake was included. The right sinonasal/maxillary bulky ellipsoid improved from second physically to first biologically. Five cases acquired model-derived OAR-adjacent burden flags, all involving brainstem-adjacent regions. These flags indicate divergence between physical dose metrics and model-derived biological scores; they do not imply clinical brainstem toxicity.

3.4 Vascular uptake was measurable but secondary

The anatomical vascular sink generally reduced effective burden relative to the no-sink biological mode, but the effect was smaller than the overall shift caused by adding non-local signalling. PTV D_{95}, peri-GTV spill, brainstem-adjacent score and parotid mean were all lower with anatomical sink than without sink, but remained above the physical-only interpretation. Therefore, the sink acted as a modifier of propagated burden rather than a mechanism that restored physical-dose interpretation.

Repeat-run uncertainty supported this interpretation (figure 6). Across evaluated case-endpoint comparisons, 40/42 sink-driven endpoint deltas exceeded the combined 95% uncertainty band. However, no case showed a noise-qualified sink-driven rank change in the repeated subset. Thus, vascular uptake was reproducible at the endpoint level but weak as an independent rank-reversal mechanism.

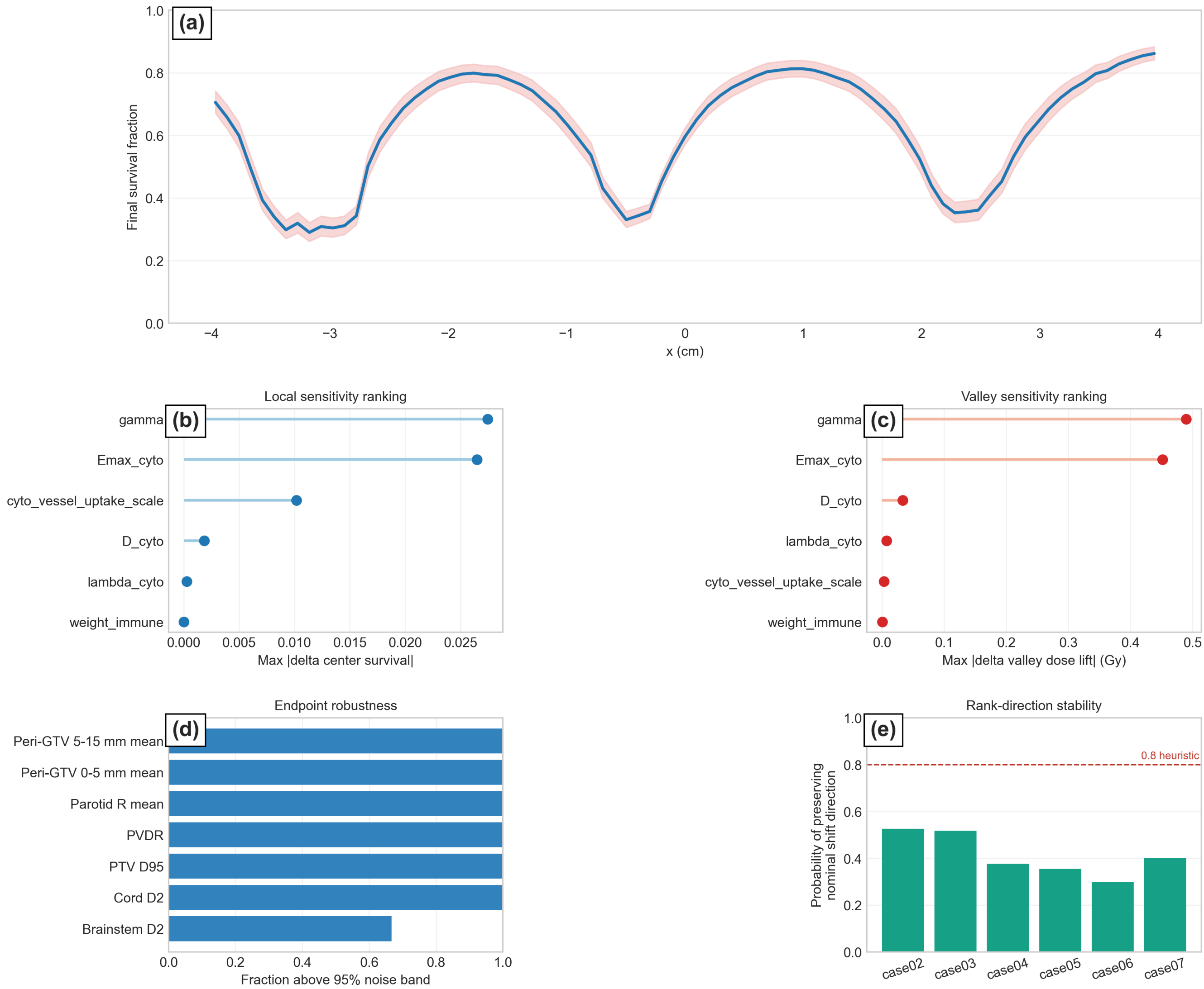

Figure 6: Uncertainty and sensitivity of the biology-informed lattice risk-analysis framework. (a) Centerline final survival with the local uncertainty band. (b) Parameter ranking for center-survival sensitivity. (c) Parameter ranking for valley effective-dose shift. (d) Endpoint-level robustness of sink-related effects, expressed as the fraction of comparisons exceeding the combined 95% Monte Carlo and uptake-sensitivity noise band. (e) Rank-direction stability across the repeated subset, with the dashed line indicating the 0.8 stability heuristic.

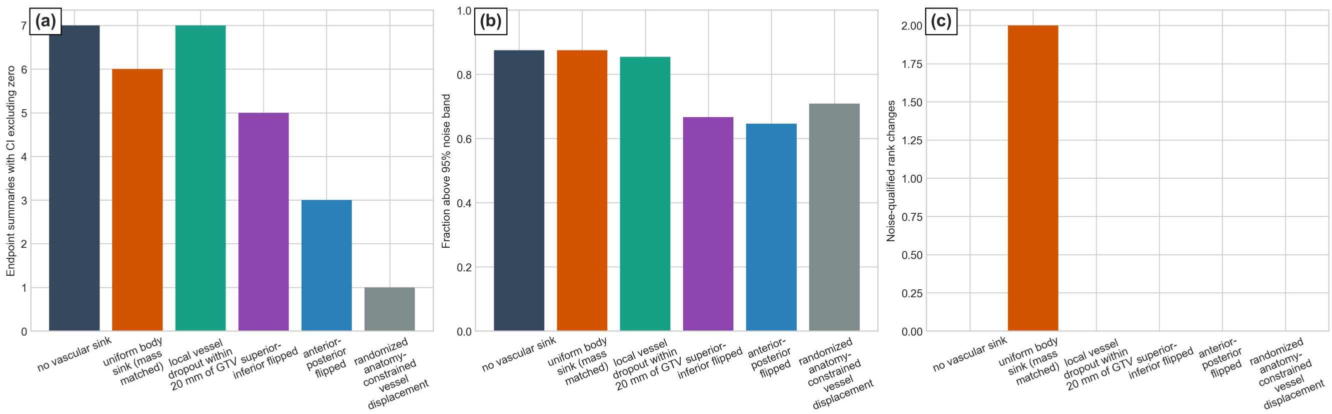

Figure 7: Falsification analysis of the vascular sink model. (a) Baseline separation between the anatomically placed sink and each comparator, expressed as the number of endpoint summaries with confidence intervals excluding zero. (b) Endpoint-level robustness in the repeated subset, expressed as the fraction of comparisons exceeding the combined 95% Monte Carlo and uptake-sensitivity noise band. (c) Rank-level impact of each falsification model, shown as the number of noise-qualified rank changes relative to the anatomically placed sink.

Table 3: Vascular-sink falsification in the repeated subset. Endpoint differences are true anatomical-sink versus comparator differences exceeding the combined 95% noise band.

Comparator	Endpoints above 95% band	Noise-qualified rank changes
No sink	42/48	0/6
Uniform body washout	42/48	2/6
Local vessel dropout within 20 mm	41/48	0/6
Randomized anatomy-constrained displacement	34/48	0/6
Superior-inferior flip	32/48	0/6
Anterior-posterior flip	31/48	0/6

3.5 Cohort-level assay-like outputs supported a propagated valley-burden interpretation

Assay-like readouts were extracted across the 10-case synthetic head-and-neck cohort to connect the endpoint-level reinterpretation to experimentally recognizable biological patterns. These outputs were treated as relative in silico proxies, not calibrated laboratory measurements.

Across the cohort, valley-region γH2AX increased from 0.335 in the physical-only analysis to 0.526 in both biology-informed modes. Peak-region γH2AX changed more modestly, from 0.626 physically to 0.682 without vascular uptake and 0.681 with anatomical uptake. A similar pattern was observed for the TUNEL-like apoptosis proxy. Valley-region TUNEL increased from 0.335 physically to 0.894 without uptake and 0.873 with anatomical uptake, while peak-region TUNEL was already near saturation after biological reinterpretation, reaching 0.991 without uptake and 0.975 with anatomical uptake.

The largest separation occurred in the cytokine-related outputs. Global cytokine AUC increased from 0 in the physical-only condition to 145.984 without vascular uptake and 135.516 with anatomical uptake. Valley-region cytokine AUC increased from 0 physically to 1620.587 without uptake and 1467.495 with anatomical uptake. Anatomical vascular uptake therefore modestly reduced cytokine and valley-burden signals relative to the no-sink condition, but did not eliminate the propagated biological burden.

These cohort-level assay-like outputs support the interpretation that biological reinterpretation was driven primarily by propagated non-local signalling in valley and peri-tumour regions, rather than only by increased injury inside high-dose vertices. The relatively modest effect of anatomical vascular uptake on peak-region injury, compared with its clearer influence on cytokine-related and valley-region outputs, supports the interpretation that the sink primarily modulates diffusible bystander burden rather than direct hotspot injury.

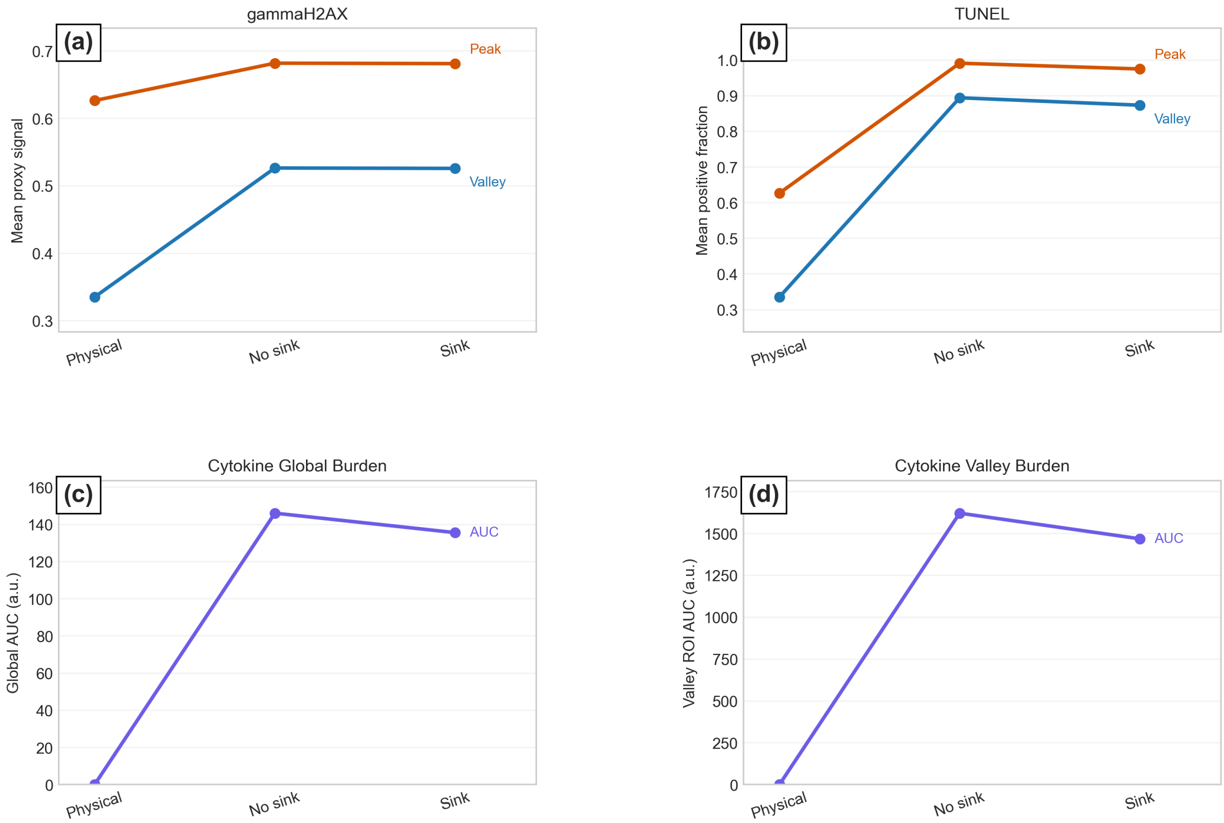

Figure 8: Assay-like biological readouts. (a) Mean gammaH2AX proxy signal in peak and valley regions across physical-only, no-sink, and anatomical-sink conditions. (b) Mean TUNEL proxy response in peak and valley regions across the same modes. (c) Global cytokine burden expressed as area under the curve. (d) Valley-region cytokine burden expressed as area under the curve.

4 Discussion

The principal finding is that physical valley dose is not equivalent to model-derived biological valley burden. Across calibration geometries and the synthetic head-and-neck cohort, biology-informed post-processing altered the interpretation of physically plausible lattice dose distributions. This is consistent with SFRT literature cautioning against interpreting treatment response using peak dose or physical heterogeneity alone Fernandez-Palomo et al. (2022); Prezado et al. (2024). The observed increase in valley and peri-tumour burden aligns with evidence that valley dose and spatial scale can be critical response determinants rather than secondary descriptors.

The biological mechanism is consistent with the established radiation-induced bystander-effect literature. Bystander responses occur in cells receiving little or no direct radiation dose and may be mediated by soluble factors, oxidative stress, inflammatory cytokines and intercellular communication pathways Prise and O'Sullivan (2009); Seymour and Mothersill (2004); Azzam et al. (2003); Blyth and Sykes (2011). The ROS-like and cytokine-like fields used here do not reproduce every molecular pathway. They provide a tractable representation of the central concept that radiation injury can propagate beyond the spatial region of direct energy deposition, explaining why biological maps became broader and more connected than the physical peak layout.

The result that non-local biology can reshape lattice-plan interpretation is supported by previous modelling. Ebert et al. separated direct and bystander response, McMahon et al. developed a kinetic intercellular-signalling model, and Peng et al. showed that bystander models can improve prediction in gradient fields compared with classical LQ-only models Ebert et al. (2010); McMahon et al. (2013); Peng et al. (2018). More directly, Cahoon et al. modelled spatial fractionation and bystander effects, supporting the idea that cells in low-dose regions may experience meaningful biological effects despite reduced physical dose Cahoon et al. (2021). The present work extends that concept to lattice-style geometries, synthetic head-and-neck anatomy and explicit vascular-sink falsification.

The vascular-sink results should be interpreted as a secondary anatomy-dependent modulation rather than the dominant driver. The largest shift arose from adding non-local signalling; vascular uptake then modified this burden according to anatomy. Prior SFRT and microbeam work suggests that vascular response, redistribution, perfusion and endothelial effects may contribute to differential tumour and normal-tissue response Bouchet et al. (2015); Ortiz and Ramos-Méndez (2024), and SFRT has been discussed as having immune and microenvironmental consequences beyond direct DNA damage Lu et al. (2024). The present sink term therefore has a defensible biological rationale, but remains a simplified clearance model rather than a full haemodynamic description.

A key strength is that the vascular-sink term was falsification-tested. The anatomical sink remained distinguishable from no-sink, uniform washout, spatial perturbation and local vessel-dropout controls at the endpoint level. This supports the interpretation that the sink is not merely a smoothing term or generic washout field. However, sink-driven rank reversals were uncommon after uncertainty was included. This useful positive-negative result suggests anatomical uptake is measurable and spatially informative, but not strong enough in this model to function as a robust independent plan-selection mechanism.

The model-derived brainstem-adjacent burden flags require careful interpretation. They are not toxicity predictions and should not be read as evidence that the synthetic plans would produce clinical brainstem injury. The brainstem values reported after biological reinterpretation are effective-dose-like scores produced by the non-local model, not absorbed dose and not normal-tissue complication probabilities. Their purpose is to identify regions where physical dose metrics and model-derived biological burden diverge. The appropriate conclusion is that non-local modelling can reveal potential OAR-adjacent liabilities not visible in physical dose alone, not that it currently predicts real brainstem toxicity.

The uncertainty and sensitivity results support a disciplined interpretation. Morris-type screening indicated that emission saturation and cytokine persistence were stronger drivers of valley-response variability than immune weighting, consistent with the purpose of elementary-effects sensitivity analysis in computational models Morris (1991); Herman and Usher (2017). Future experimental calibration should therefore prioritize signal generation and persistence parameters. The assay-like outputs also provide a bridge to experimental radiobiology: γH2AX relates to DNA double-strand-break signalling and TUNEL to apoptotic DNA fragmentation Rogakou et al. (1998); Gavrieli et al. (1992). These outputs are not calibrated measurements, but express predictions in experimentally recognizable terms.

Several limitations define the study scope. First, the framework is entirely synthetic and hypothesis-generating: phantoms, tumour geometries, vessel masks and lattice placements were algorithmically generated. Second, the physical deliveries were TOPAS-derived lattice surrogates rather than TPS-optimized clinical plans. They reproduce broad lattice-planning features, but should not be interpreted as deliverable treatments. Third, Monte Carlo history counts were intentionally modest to enable repeated experiments across cases and uncertainty conditions. Although repeated seed/history analyses and dose-standard-deviation outputs were used, higher-statistics transport and convergence maps would be required before translation. Fourth, the vascular sink was static and did not model flow direction, perfusion heterogeneity, permeability, endothelial injury, immune-cell trafficking or time-varying transport. Fifth, biological outputs are model-derived scores and relative proxies, not experimentally validated survival, cytokine, DNA-damage or apoptosis measurements.

Despite these limitations, the framework provides a useful stress test for lattice plan interpretation. Much of the lattice-radiotherapy literature focuses on delivery, vertex geometry, target coverage, PVDR and OAR sparing Duriseti et al. (2021); Kavanaugh et al. (2022); Duriseti et al. (2022); Grams et al. (2023); Zhang et al. (2023); Botti et al. (2024). The present study asks a different question: whether the meaning of a physically acceptable lattice pattern changes when non-local signalling and anatomy-aware uptake are considered. The results suggest that physical DVH-style interpretation alone may miss biological liabilities in valley, peri-tumour and organ-adjacent regions. The next step is validation using patient-derived RTSTRUCT/RTDOSE exports, realistic vascular segmentation and partial-field or lattice-like assay data. Until then, the framework should be viewed as a computational risk-analysis tool, not a clinical predictor.

5 Conclusion

A locked reaction–diffusion bystander model combined with synthetic head-and-neck lattice cases showed that non-local biological signalling can materially alter the interpretation of physically plausible SFRT plans. Anatomy-aware vascular uptake produced smaller but measurable endpoint modulation and remained distinguishable from surrogate washout controls. The framework is not a validated clinical decision tool, but provides a compact, testable risk-analysis layer for identifying biological liabilities not captured by physical dose metrics alone.

Declarations and data availability

Acknowledgements: none at this time. **Funding:** no financial support was received for this work. **Conflicts of interest:** the author declares no conflicts of interest. **Ethics:** this study used only synthetic computational phantoms and involved no patient-identifiable data, human participants or animal experiments. **Data and code availability:** synthetic phantom definitions, lattice-case metadata, TOPAS templates, post-processing scripts, biological model code, endpoint extraction routines and figure-generation scripts should be made available in a public repository at submission or publication, with a permanent archived version deposited in Zenodo or an equivalent repository. For peer review, the repository should include the synthetic case manifest, mask-generation scripts, raw or down-sampled dose grids where file size permits, repeated seed/history uncertainty outputs, biological parameter files and scripts required to reproduce all main figures and tables. Repository placeholder: `https://github.com/kair98-boop/SFRT`. Archived DOI placeholder: Zenodo DOI to be added before final submission.

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